

RetailPulse — Data Cleaning Notes

1. Dataset Overview

The original dataset contained **1,825 order records and 11 columns**, covering the period from January to June 2026.

The main data-quality issues identified were missing values, duplicate records, inconsistent city formatting, and unusual numerical values.

2. Issues Identified and Treatment

Issue	What we found	What we did
Duplicate records	25 duplicate rows were found.	Removed the 25 duplicate rows using <code>drop_duplicates()</code> , reducing the dataset from 1,825 to 1,800 rows.
City formatting	City names had inconsistent capitalization such as <code>Mumbai</code> , <code>mumbai</code> , <code>MUMBAI</code> , and <code>Bengaluru/Bangalore</code> .	Removed extra spaces, standardized capitalization using <code>.str.title()</code> , and changed <code>Bangalore</code> to <code>Bengaluru</code> so each city had one consistent name.
Missing quantity	36 quantity values were missing.	Missing quantities were filled using the median quantity, so the affected orders could still be included in the analysis.
Missing unit price	57 unit-price values were missing.	Missing prices were filled using the median unit price, allowing revenue to be calculated without dropping those orders.
Missing discount	37 discount values were missing.	Missing discount values were replaced with 0%, treating them as orders with no recorded discount.
Missing payment method	27 payment methods were missing.	Replaced missing values with <code>Unknown</code> rather than guessing the customer's payment method.
Missing customer age	74 customer-age values were missing.	Missing ages were filled using the median customer age to retain the orders for analysis.

Missing customer rating	529 ratings were initially missing; 520 remained missing after cleaning.	Ratings were left blank because a missing rating cannot reliably be replaced with an assumed score.
Unusually high unit prices	Extremely high prices were found for Backpack, Electric Kettle, Sunscreen SPF50 and Honey Jar.	Compared these values with the typical price for each product. Clear outliers were replaced with the median price of the same product rather than deleting the orders.
Unusually high quantities	Four orders had a quantity of 999, which was far outside the normal range.	Treated these as likely data-entry errors and replaced 999 with the median quantity for the corresponding product.
Invalid numerical values	Checks were performed for negative quantities/prices, discounts outside 0–100%, and ratings outside 1–5.	No invalid values remained after cleaning: 0 negative quantities, 0 negative prices, 0 invalid discounts and 0 invalid ratings.

3. Outlier Investigation

The unusually high unit prices were not removed automatically. They were compared with the typical prices of the same products.

The extreme values were substantially higher than their respective product medians, so they were treated as likely data-entry errors and corrected using product-level median prices.

Similarly, four records had a quantity of **999**. Since the normal product quantities were much lower, these values were treated as data-entry errors and replaced using the median quantity for the relevant product.

4. Final Validation

After cleaning, the dataset contained **1,800 orders and 13 columns**. Two additional columns, revenue and day_type, were created for the analysis.

The final validation showed:

- **0 duplicate rows**
- **0 negative quantities**
- **0 negative prices**
- **0 invalid discount values**
- **0 invalid customer ratings**
- **520 missing customer ratings**, which were intentionally retained

5. Result

The cleaned dataset was saved as **retailpulse_orders_cleaned.csv** and was used as the basis for the revenue, weekday/weekend, discount and product-level analysis.